Final Project: Pick a Project

Computer Science - Week 10

Oct 3, 2026 - Version 0.0.2

Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Anthony DeMare	apdemare@udel.edu
Friend (optional)	Soumio	

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea

and a paragraph describing how that paper supports your design concept.
        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider the project below.
        - Create an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the chempy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.

As a mathematics major, I'm probably not going to do something with math since so many things have complicated simplifications. Therefore, I'm either going to attempt a derivatives practice website, an escape room type of website, or I'll create a poker game based on the previous poker project, expanding it to include blackjack, betting amounts, and different variations of poker.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

There are many derivative practices, escape rooms, and many many poker websites.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

If I continue with the derivatives, it would be a good practice for people learning derivatives. The other two ideas would simply be for some quick entertainment.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

I'll use the poker example. The website will take an input from a select box, either Texas hold'em, five card, three card, or blackjack, and begin a game based on the selection. Next the program would generate a hand, secretly generate a dealer hand, and reveal any necessary information. The website will then take the input from the player (hit, split, fold, call, raise) and execute functions based on the input. It will then calculate results based on hands, add or subtract chips, and prompt the next round.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

Nothing specific besides choosing from a preset list of options.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

The computer will only need to add or subtract chip amounts, but will need to calculate poker hand scores against each other to determine the winner of rounds.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The user will get a final chip amount and either a win or a loss.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things"

(nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Hands, cards, chips, players.

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Cards: Card suit, card rank, in hand or community, whose hand.

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

The score calculation and figuring out when the opponent should fold.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

It will not provide code, just ideas or suggestions with vague hints like “you could use a loop.”

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

My goal for a poker website would be to make an easy to use website that provides enjoyment for the player and accurate representation of the poker playing experience.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to build a simple web tool to run a poker game simulation. What are 3 simple features a basic tool should have?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

The AI suggested card dealing and shuffling, hand evaluation and winner detection, and game setup and simulation controls. They did not mention any existing solutions or previous apps.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

I feel like developing a shuffling program would be difficult, could it just be a random number and suit generator and then check subsequent cards for duplicates?

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

I now can use two random number generators and simply reroll if there are duplicates, probably using a “while card in [cards]”.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

I will reference previous code, like the already created three card poker.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

I will check what lines in the code return an error and employ any bug fixes.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I want to build a simple web tool to run a poker game simulation. What are 3 simple features a basic tool should have?
For a simple poker game simulation, I'd start with these three features:

1. Card Dealing and Shuffling

Create a standard 52-card deck.

Shuffle the deck randomly.

Deal two cards to each player and five community cards for Texas Hold'em.

2. Hand Evaluation and Winner Detection

Evaluate each player's hand, from a high card to a royal flush.

Compare hands to determine the winner.

Display the winning hand and the players who won or tied.

3. Game Setup and Simulation Controls

Let users choose the number of players, such as 2 to 6.

Include a button to start a new round.

Display each player's cards, the community cards, and the round's results.

Recommended development order: Build card dealing first, add hand evaluation second, then create the game interface and controls. This keeps the project manageable while giving you a working simulation.

I feel like developing a shuffling program would be difficult, could it just be a random number and suit generator and then check subsequent cards for duplicates?
Yes! That would be a perfectly reasonable approach for a basic poker simulation. You don't need to build a complicated shuffling algorithm.

You could generate each card randomly and check whether another player already has the same card.

How it would work
Generate a random number from 2 to 14, where 11 = Jack, 12 = Queen, 13 = King, and 14 = Ace.

Generate a random suit from hearts, diamonds, clubs, and spades.

Check for duplicates. If the generated card already exists in the current round, generate another card. Otherwise, add the card to the game.

Repeat until every player has two cards and the five community cards have been dealt.

For example:

Example of duplicate checking

Robot or human?

Generated card: Ace of Spades

Unique
Add to the game.

Robot or human?

Generated card: Ace of Spades

Duplicate
Reject it and generate another card.

One improvement to consider
Your method works, but checking for duplicates becomes less efficient as the deck fills up. For a standard 52-card deck, this is a minor issue.

My recommendation: Use your approach for a beginner-level project. Later, if you want to simplify the code further, generate all 52 unique cards in a list and use Python's shuffle() function. That handles duplicates automatically because every card already appears only once.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.